Capstone: Catalogues and Your Own Observing Plan

Covers Lecture 3.5 and brings the whole course together 2–3 weeks

Name: _____ Date: _____ Marks: _____ / 80

The idea

Real astronomers don't just "go out and look" — they **plan**: what is up, when, how high, with which instrument and which eyepiece. In this project you plan **and carry out** a real observing night. You need Stellarium (or an atlas), the observing list from Lecture 3.5, the seasonal sky charts, and your telescope or binocular specifications (or use Telescope X from Assignment 3B: 130 mm, $F = 650$ mm, eyepieces 25/10/6 mm).

Part A — Catalogue knowledge 20 marks

A1. Match the object to its type (½ each, 4 marks)

Object	Type
1. M1 (Crab Nebula)	A. Globular cluster
2. M13	B. Spiral galaxy
3. M31	C. Emission nebula
4. M42	D. Open cluster
5. M45	E. Planetary nebula
6. M57	F. Supernova remnant
7. NGC 5139 (Omega Centauri)	G. Globular cluster
8. NGC 869/884	H. Open cluster (double)

A2. Who made it? (1 each, 6 marks) (a) Messier catalogue _____ (b) NGC _____ (c) Caldwell _____
(d) main classifier for the Henry Draper spectral types _____ (e) the 1.8-billion-star catalogue (mission) _____ (f) the first star catalogue (~130 BCE) _____

A3. Short answers (2 each, 10 marks)

1. Why did Charles Messier make his list?
2. How do **open** and **globular** clusters differ (age, number of stars, location in the galaxy)?
3. A planetary nebula has nothing to do with planets. What is it?
4. Why is a galaxy of magnitude 9 often harder to see than a star of magnitude 11?
5. Why is India a good place for a Messier Marathon, and in which month should it be attempted?

© Vaibhav Yenare

Part B — Can you see it from here? ★★ 12 marks

Use $h_{\text{transit}} = 90^\circ - \varphi + \delta$; objects lower than $\sim 15^\circ$ are usually lost in haze. (½ per altitude + ½ per decision.)

Object	Dec	Delhi (28.6° N)	Mumbai (19.1° N)	Bengaluru (13° N)	Worth trying from Delhi?
Omega Centauri	−47.5°				
47 Tucanae	−72.1°				
Carina Nebula	−59.9°				
M7	−34.8°				
M13	+36.5°				
M81	+69.1°				

Part C — Plan your observing night ★★ 28 marks

C1. Choose the night (4). Date (within ±4 days of Amavasya — why?) · sunset and end of astronomical twilight (~75–80 min after sunset) · Moon rise/set · weather forecast · site and its latitude/longitude.

C2. Pick 8 targets (16; 2 per complete row). Choose objects above 25° during your session: at least 1 planet or the Moon, 2 star clusters, 1 nebula, 1 galaxy, 1 double star and 2 of your choice.

#	Object (ID & name)	Type	RA	Dec	Mag	Altitude	Star-hop from	Eyepiece & M	Fits field?
1									
2									
3									
4									
5									
6									
7									
8									

C3. Order them (4). Arrange your targets in observing order. Rule of thumb: **observe objects in the west first** (they set soon). Explain how RA helps you decide.

C4. Risk plan (4). What will you do if (a) clouds cover half the sky, (b) the finder is misaligned, (c) a target is too faint from your site?

Part D — Observe and log ★★ 20 marks

At the course star party (or with your own instrument) observe **at least 4** of your targets. For each, log the details and make a pencil **sketch** (stars as dots, nebulae as soft shading; note the field width and orientation).

Object	Time	Instrument & M	Seeing (1–5)	Transparency (1–5)	What I saw

Reflection (6 of the 20): which object surprised you most, and how did it compare with photographs? Did your calculations (field of view, magnification, limiting magnitude) match what you experienced? (Log 8 · sketches 6 · reflection 6)

Part	A	B	C	D	**Total**
Marks	20	12	28	20	**80**